

Genetics of common complex psychiatric disorders I

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Part 1: Biometrics

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Genetics and Environmental Influences on Behaviour and Mental Health

What is a “common”, “complex” psychiatric disorder?

Common: Affects 1% or more of the population

Complex: Inheritance cannot be explained by a single gene

Psychiatric disorders are defined by disruption to higher-order brain functions of moods, perceptions, thoughts, beliefs, and behaviours but usually in the absence of major neurological impairments (consciousness, senses, memory). Psychiatric disorders include depressive and anxiety disorders (major depressive disorder, panic disorder), manic and psychotic disorders (bipolar disorder, schizophrenia), obsessive-compulsive disorders, eating disorders (anorexia nervosa, bulimia nervosa), substance-use disorders and personality disorders. Childhood conditions like attention-deficit/hyperactivity and autism can also be included, but only when they lead to clinically-salient impairment or distress. There are also many shades of sadness, hallucinations, eccentricities, mood swings, body-image preoccupations, recreational substances use, personalities, etc that are not psychiatric disorders but may still be informative to study from an aetiological and genetic standpoint.

- Sullivan PF and Geschwind DH (2019) [Defining the Genetic, Genomic, Cellular, and Diagnostic Architectures of Psychiatric Disorders](https://doi.org/10.1016/j.cell.2019.01.015). *Cell* doi:10.1016/j.cell.2019.01.015

Psychiatric diagnoses

- Depressive disorder: Marked and persistent low mood and inability to feel pleasure.
- Anxiety disorder: Extensive, pervasive, unrealistic, and disabling worry.
- Bipolar disorder: Intense mood swings between mania and depression.
- Schizophrenia: Persistent hallucinations and delusions that severely impair functioning, highly disorganised thought and speech.
- Eating disorders: self-starvation or excessive over-eating, debilitating preoccupation with body image.
- Attention-deficit/hyperactivity disorder: Early onset, developmentally inappropriate and persistent hyperactivity and inattention.
- Autism spectrum disorders: Markedly impaired interpersonal interactions, non-goal-directed behaviours, and language development.
- Sullivan, P. F., Daly, M. J. & O'Donovan, M. Genetic architectures of psychiatric disorders: the emerging picture and its implications. *Nat Rev Genet* 13, 537–551 (2012).
- Frances, A. *Essentials of Psychiatric Diagnosis*. Guilford Press (2013).

Prevalences

- Depression: 3% in a week
- Schizophrenia: 1% in lifetime
- Bipolar disorder: 2% in lifetime
- Anxiety disorder: 6% in a week

Psychiatric disorders have many causes, correlates, and consequences (genetics, environment, family life, substance use, relationships)

Incidence of psychiatric disorders range from the common (depression, anxiety) to the rare (bipolar disorder, schizophrenia).

Why genetics?

Why use genetics to study mental health and psychiatric disorders?

- Biological understanding of genes, pathways
- Shared aetiology with other disorders
- Risk prediction
- Drug retargeting
- Causal analysis of environmental risk factors

Genetics of categorical traits

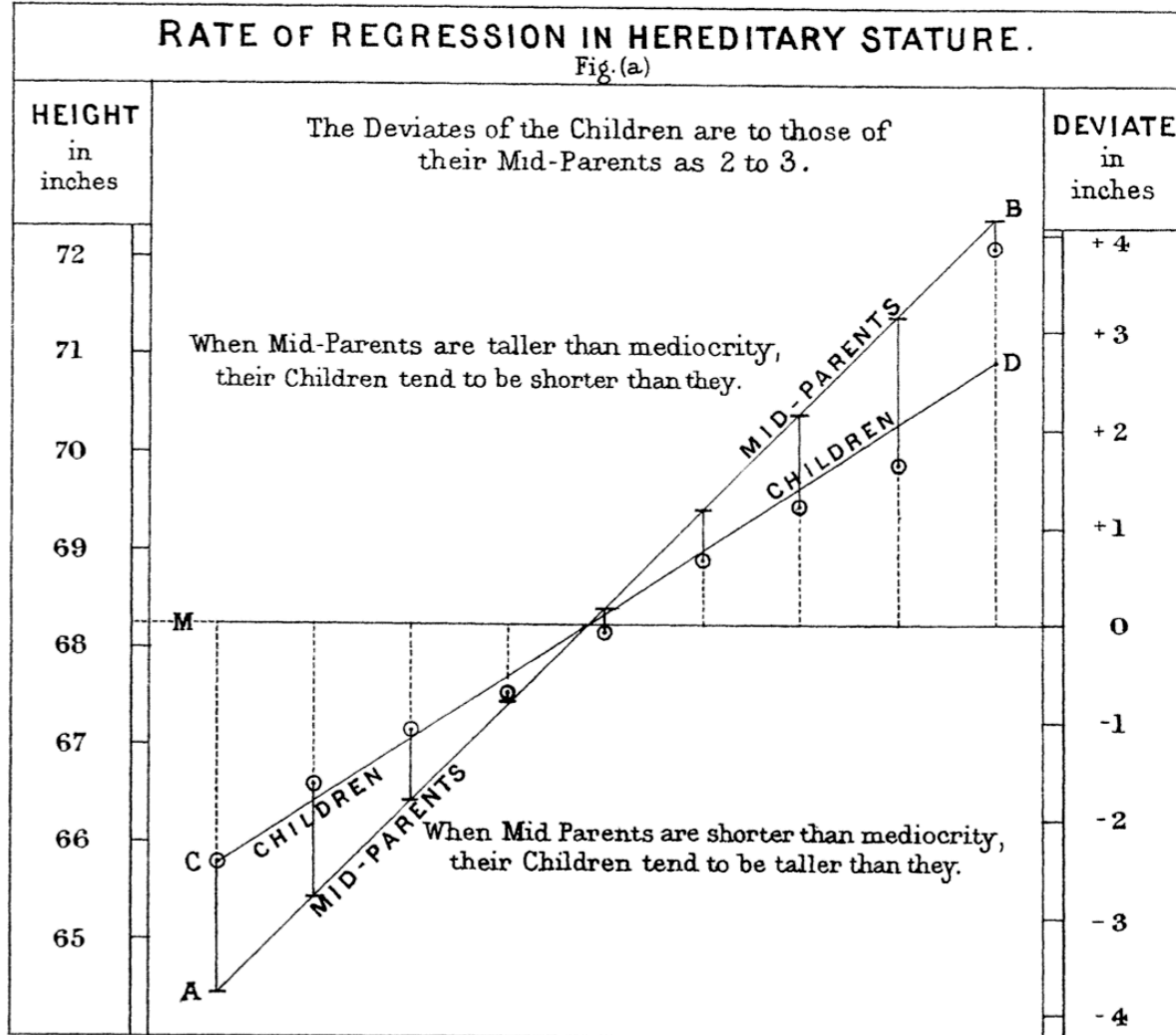
Seed		Flower	Pod		Stem	
Form	Cotyledons	Color	Form	Color	Place	Size
						
Grey & Round	Yellow	White	Full	Yellow	Axial pods, Flowers along	Long (6-7ft)
						
White & Wrinkled	Green	Violet	Constricted	Green	Terminal pods, Flowers top	Short (1/2 - 1ft)
1	2	3	4	5	6	7

Figure 1: Diagram showing the seven “characters” observed by Mendel

Gregor Mendel (1822–1884), working in what is now Czechia, discovered the transmission of traits from parents to offspring could be explained by the inheritance of two “elements”, which we now call alleles. Mendel was concerned with discrete or categorical phenotypes.

[Mendel pea plant figure](#) by Mariana Ruiz (LadyofHats) [public domain]

Genetics of continuous traits

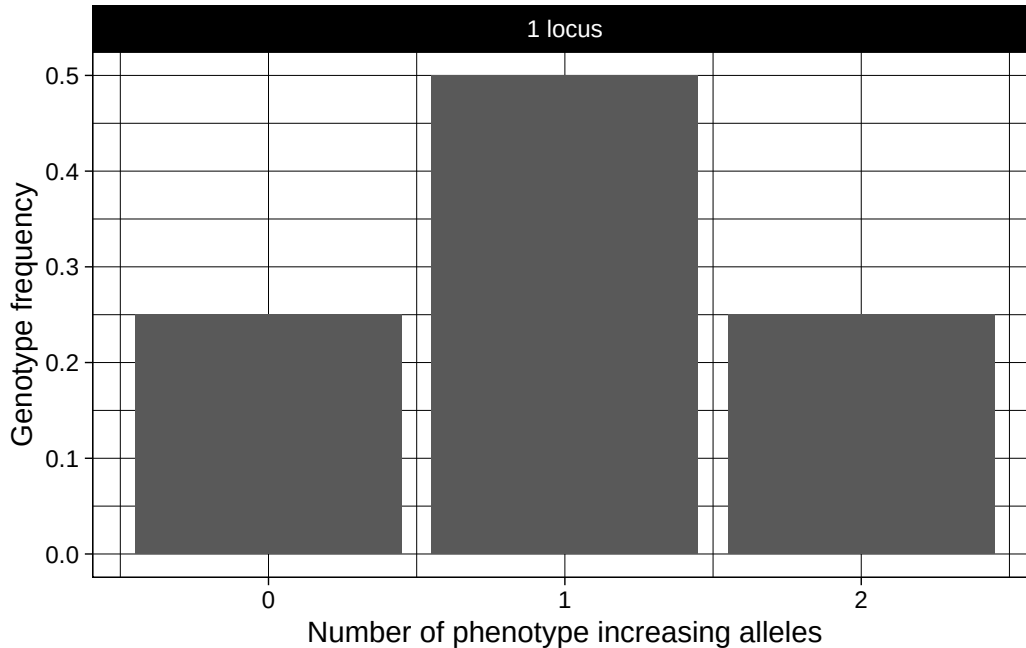


Separately, Francis Galton (1822–1911), was studying the inheritance of continuous or metric phenotypes. He noticed the parents who were tall tended to have children that were slightly shorter than themselves (and vice versa). This was termed “regression to the mean” from which the name of the statistical method “regression” is derived.

“In their search for universal hereditary laws, Galton and Pearson were driven by the linear model and the normal distribution because the associated parameters had scientific meaning for them that went beyond mere description.” - Wachsmuth, A., Wilkinson, L., & Dallal, G. E. (2003). [Galton's Bend](#). *The American Statistician*, 57(3), 190–192. doi:10.1198/0003130031874

For more on Galton's legacy, see <https://adelphigenetics.org/history/>

Reconciling categorical + continuous genetics = quantitative genetics



Ronald Fisher reconciled the inheritance of continuous and categorical phenotypes by showing that a continuous phenotype could be made from the inheritance of a large number (dozens, hundreds, or thousands) categorical genes. The term “variance” comes from Fisher’s discoveries.

- Fisher, R. A. (1918). [XV.—The correlation between relatives on the supposition of Mendelian inheritance.](#) *Transactions of the Royal Society of Edinburgh*. doi:10.1017/S0080456800012163
- Charlesworth and Edwards (2018). [A century of variance.](#) *Significance* 15(4). doi:10.1111/j.1740-9713.2018.01170.x
- Bodmer et al (2021) [The outstanding scientist, R.A. Fisher: his views on eugenics and race.](#) *Heredity* doi:10.1038/s41437-020-00394-6

Single locus with additive effect

Single locus with two alleles, A_1 and A_2 , and four genotypes with values:

- A_1A_1 : $2a$
- A_1A_2 and A_2A_1 : a

- A_2A_2 : 0

At this locus the A_2A_2 homozygote is set as the baseline

If allele A_1 has a frequency of p and A_2 has frequency $1 - p$, then the genotype frequencies are:

- A_1A_1 : p^2
- A_1A_2 and A_2A_1 : $2p(1 - p)$
- A_2A_2 : $(1 - p)^2$

Genotype	Frequency	Value	Freq. $\times$ Val.
A_1A_1	p^2	$2a$	$2p^2a$
A_1A_2	$p(1 - p)$	a	$p(1 - p)a$
A_2A_1	$p(1 - p)$	a	$p(1 - p)a$
A_2A_2	$(1 - p)^2$	0	0
		Mean =	$2pa$
		Variance	$2p(1 - p)a^2$
		=	

Table based on Table 7.1, Falconer and Mackay.

Genetic values

Each individual i has a genetic value at locus j :

$$g_{ij} \in [0, a_j, 2a_j]$$

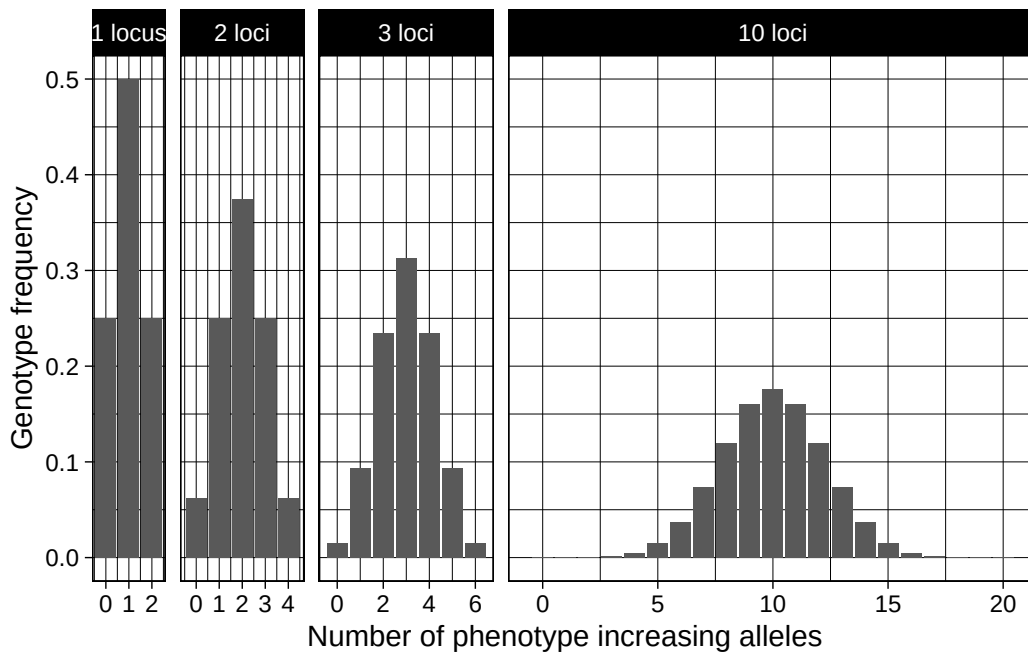
Total genetic value from M loci:

$$G_i = \sum_{j=1}^M g_{ij}$$

$$\text{var}(G) = \sum_{j=1}^M 2p_j(1 - p_j)a_j^2$$

Polygenic traits are quantitative traits

Adding up effects from a large number of genetic loci makes a continuous phenotype.



“R. A. Fisher’s 1918 paper, ‘The correlation between relatives on the supposition of Mendelian inheritance’, resolved the often bitter conflict between biometricians and Mendelians, which raged for a decade following the rediscovery of Mendel’s work. Fisher showed that a complex quantitative trait could be explained by Mendelian inheritance if several genes affect the trait.” - Plomin, R., Haworth, C. & Davis, O. [Common disorders are quantitative traits](#). *Nat Rev Genet* **10**, 872-878 (2009). doi:10.1038/nrg2670

Biometrics

What are the sources of family resemblance? How do we quantify them numerically?

Heritability

Proportion of similarity in phenotypes that can be attributed to similarity in genotypes.

Model: Phenotype (P) = Genotype (G) + Environment (E)

Variance decomposition

$$\text{var}(P) = \text{var}(G) + \text{var}(E)$$

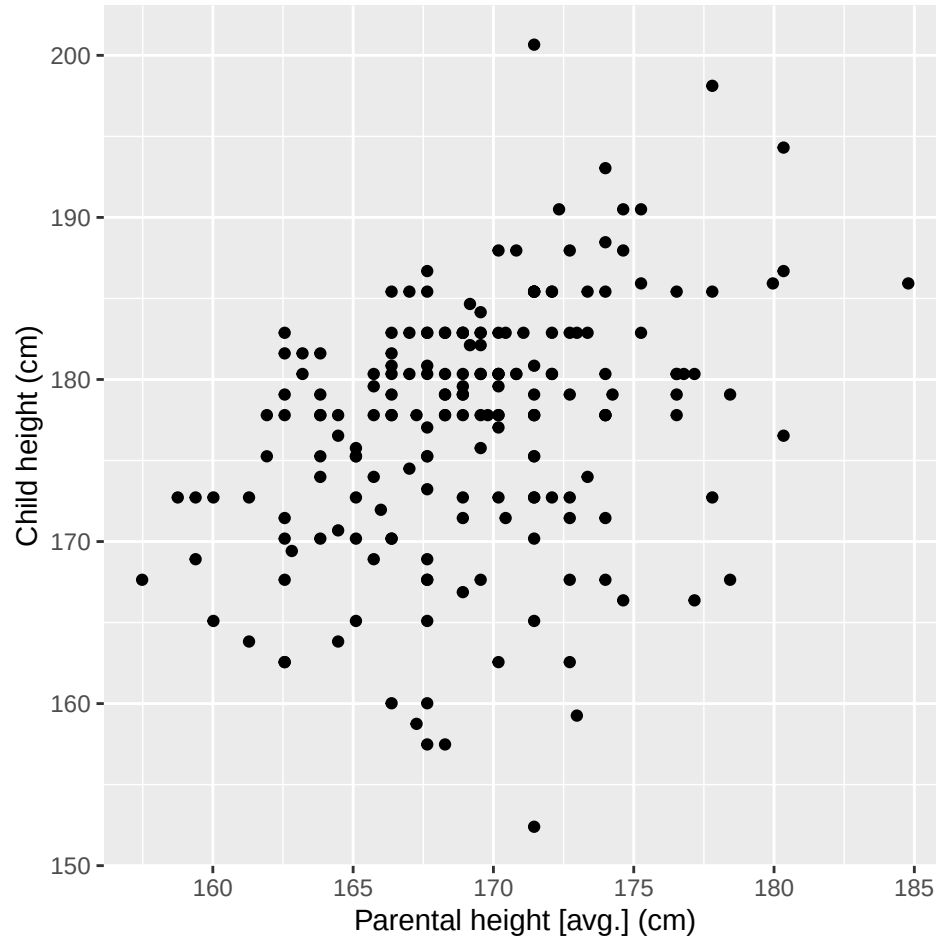
Proportion of variance

$$H^2 = \frac{\text{var}(G)}{\text{var}(P)}, e^2 = \frac{\text{var}(E)}{\text{var}(P)}, H^2 + e^2 = 1$$

The effect G denotes all genetic effects ($G = A + D + I$ for additive, dominance, and epistatic variance). In the more limited case where it is assumed genetic effects act only additively (they don't interact with each other), then only additive genetic variance $\text{var}(A)$ is used with (narrow-sense) heritability defined as $h^2 = \frac{\text{var}(A)}{\text{var}(P)}$.

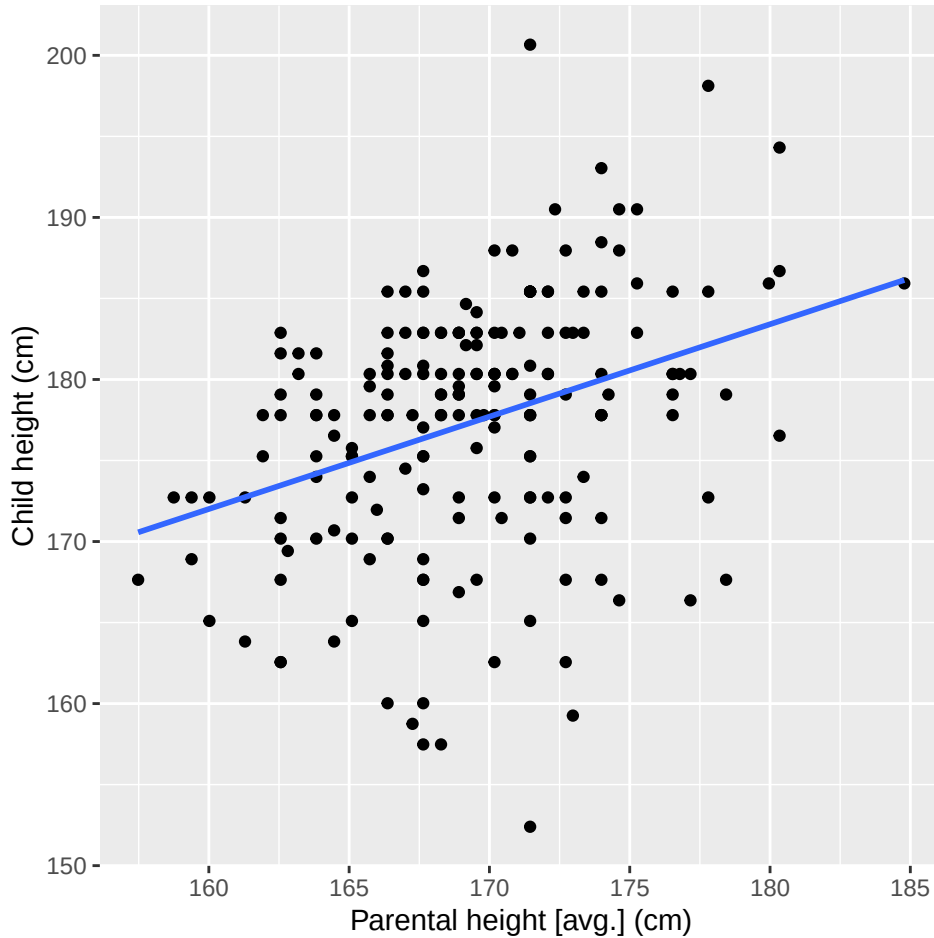
- Tenesa, A., Haley, C. [The heritability of human disease: estimation, uses and abuses](#). *Nat Rev Genet* **14**, 139–149 (2013). doi:10.1038/nrg3377
- Visscher, P., Hill, W. & Wray, N. [Heritability in the genomics era — concepts and misconceptions](#). *Nat Rev Genet* **9**, 255–266 (2008). doi:10.1038/nrg2322

How to estimate heritability from data



Plot of child (offspring) height versus the average of their parents' heights. What is a statistic that can be used to summarise the relationship between these two variables?

How to estimate heritability from data



$$\beta = \frac{\text{cov}(X,Y)}{\text{var}(X)}$$

Estimate the beta coefficient (slope) for a simple regression from the covariance between predictor (X) and outcome (Y) variable divided by the variance of the predictor (X).

Simple model of genetic and environmental effects

$$P = A + E$$

The phenotype value P is influenced by an additive genetic effect A and an environmental effect E .

For simplicity assume that P is an individual's deviation from the average phenotype in the population.

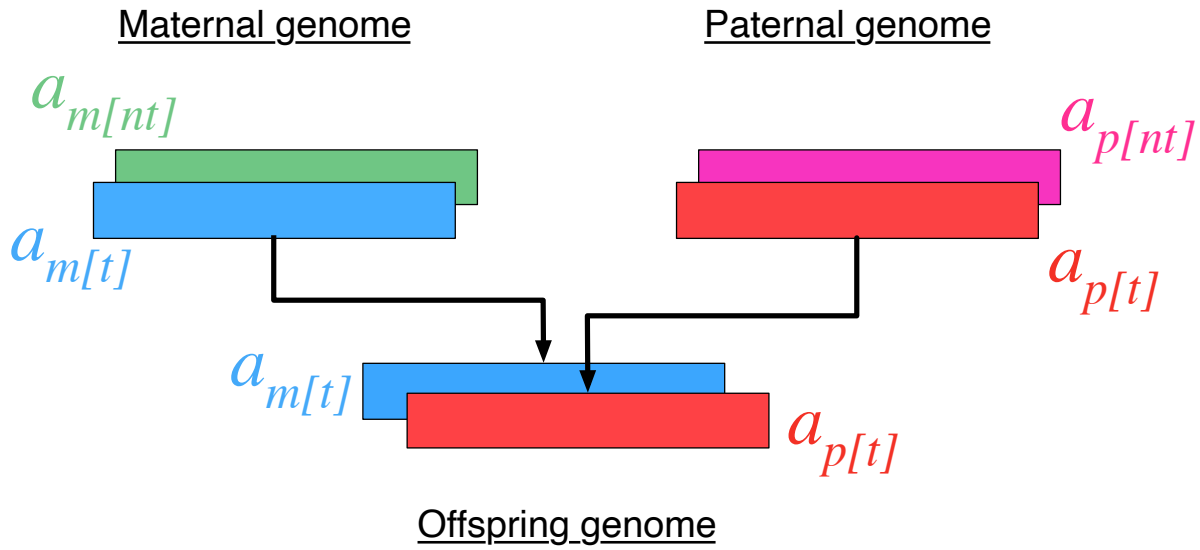
Simple model of genomics

$$A = a_{m_t} + a_{p_t}$$

Each individual has two copies of the genome, one inherited from each parent.

Here we label the genetic effect from the mother a_{m_t} (maternal) and the genetic effect from the father a_{p_t} (paternal). These are the transmitted genetic effects, denoted with a subscript t .

Simple model of inheritance



Each parent also has a copy of the genome which they do not pass on to their child (“non-transmitted”), labelled $a_{m_{nt}}$ for the mother and $a_{p_{nt}}$ for the father. Non-transmitted genetic effects are written with a subscript nt .

Simple model of genetics, environment, and inheritance

Phenotype (P) value is the sum of the two genetic values plus an environmental value (e).

- Mother’s phenotype (maternal): $P_m = a_{m_t} + a_{m_{nt}} + e_m$
- Father’s phenotype (paternal): $P_p = a_{p_t} + a_{p_{nt}} + e_p$
- Child’s phenotype (offspring): $P_o = a_{m_t} + a_{p_t} + e_o$

Regression equation

$$\beta = \frac{\text{cov}(X, Y)}{\text{var}(X)}$$

- X = average of parents' phenotypes
- Y = offspring phenotype

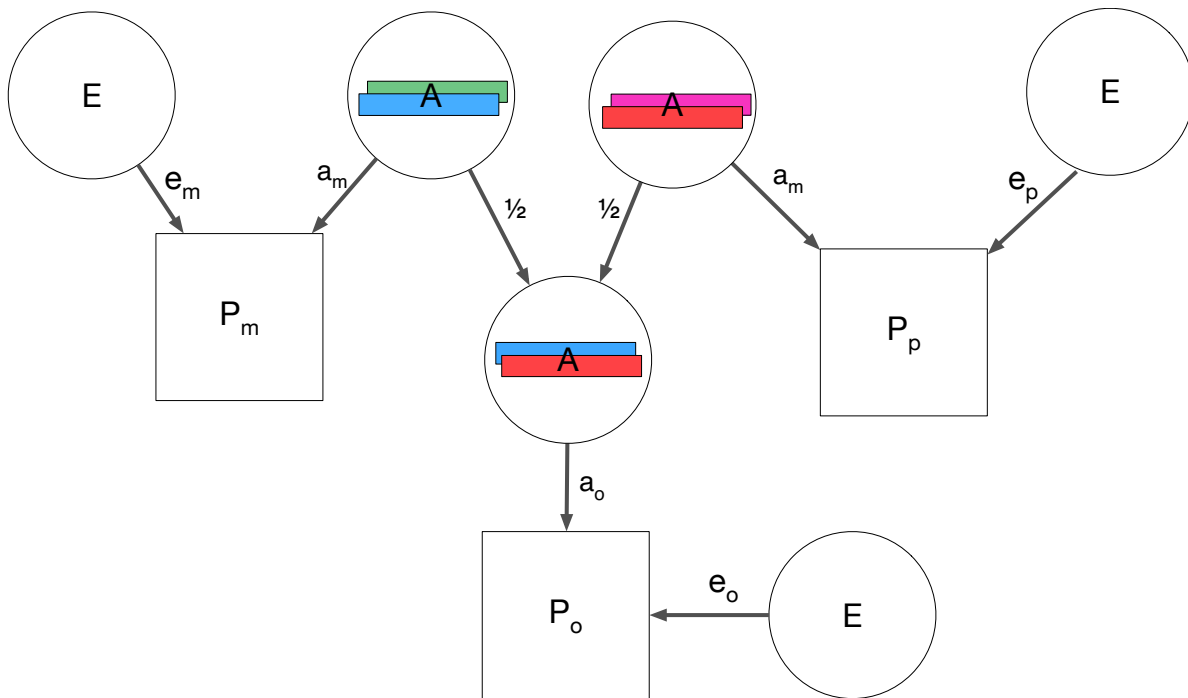
$$\text{Therefore, } \beta = \frac{\text{cov}(\frac{P_m + P_p}{2}, P_o)}{\text{var}(\frac{P_m + P_p}{2})}$$

Parent-offspring covariance

$$\text{cov}(\frac{P_m + P_p}{2}, P_o)$$

The covariance term is between the average of the parental phenotypes and the offspring phenotype.

Simple genetic model



Visualisation of the simple genetic model. The observed variables (maternal P_m , paternal P_p , and offspring P_o phenotypes) are shown as squares. Unobserved (latent variables) (genetic A